



PROFIBUS

Guideline

Profile Guidelines Part 1: Identification & Maintenance Functions

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Prepared by various PROFIBUS Working Groups mainly within the Technical Committee 3 "Application Profiles".

This series of PROFIBUS profile guidelines specifies generic profile features for all PROFIBUS devices as long as dedicated profiles do not overrule them by well-founded reasons, e.g. already existing profiles or DP-V0 functionality only.

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			New chapter 3.4 "Manufacturer specific Information"
			New chapter 3.2.1 "Header"; modified write procedure
			Data type "Visible String" 7 Bit instead of 8 Bit ASCII
			Added parameter "I&M Version" in I&M0
			Data type of parameter SIGNATURE changed to Octet String
1.1	TC3	29-Apr-2003	Ident_Maintenance_supp "optional" instead of "default"
			Reference to HCF (HART) deleted
			Hour range changed to 0...23, minutes to 0...59
			PROFILE_ID: range 0000h...00FFh changed to F600h...F6FFh
			GSD keywords within a separate chapter
			Refined chapter 7.1: Coding of IDs and Types
			Added new chapter 7.4: Rules for the Assignment of IDs ...
1.1	TC3	15-May-2003	Chapter 5.3: Error code examples for access coordination refined
			Chapter 3.3, 3.4, 4: restrictions on record lengths defined
			Chapter 1.1: rules for brandlabeled devices
1.1.1	Advisory Board	11-March-2005	Chapter 1.1, 4: MS2 services mandatory, MS1 services optional

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1 Motivation and Scope

Over the years the PROFIBUS community got acquainted with a couple of application profiles such as "PA devices", "PROFIdrive", "Encoders", and others. For these device families structures and behavior are pretty well defined and application profiles are becoming more popular. On the other hand the community is facing some flaws with profiles. One of these is the fact that it is not worthwhile for every device type to develop a profile. Thus many manufacturers are designing their devices according to their own rules and optimization strategies and hence customers are suffering from diversity in functions, syntax and semantics. Another problem comes with the staggered appearance of profiles on the market. It is nearly impossible to keep all the profiles on the same level of knowledge and standardization.

Technical Committee 3 within the PROFIBUS Organization therefore decided to publish a series of profile guidelines dealing with the most important common features of PROFIBUS slave devices. They are starting with part 1, the "Identification & Maintenance Functions" (I&M functions). The main purpose of the I&M functions described hereinafter is to support the end user during various scenarios of a device's life cycle be it configuration, commissioning, parameterization, diagnosis, repair, firmware update, asset management, audit trailing, and alike. Well-defined uniform parameters and rules should enable the manufacturers to offer devices that behave in a uniform manner and that should enable the end user to act quick and straight. These profile guidelines take in account requirements from NAMUR, the FDA (Food & Drug Administration) and others.

The guidelines are defining in its first part the "storage bin" for three kinds of I&M functions. One kind may be characterized with the term "Type Plate" or "Boiler Plate". It consists of a mandatory, an optional and a reserved area for future use. A second kind deals with profile- and a third one with manufacturer-specific parameters.

Part 2 contains a description of the parameters for the "Type Plate" and part 3 the coding of the records to be stored in and retrieved from the device. The difference between the information within the head station and within a module of a device is minor.

The I&M functions are not covering information needed for use cases like planning, bidding, ordering, adjusting, disposing, etc.

1.1 Use Cases and General Rules

The following use cases are characterizing the applications for the I&M functions within these guidelines:

- Configuration
- Commissioning (parameterisation)
- Repair and update
- Operation and visualisation (PG/PC/OP/ES/PCS/HMI/SCADA/etc.)
- Support for manufacturing execution systems (MES; "type plate")
- Asset management (minimum information for device identification)
- Audit trailing / validation (paperless, signature based; FDA-"21 CFR 11")

The following general rules apply:

- Mandatory shall be: device/module holds as little information as is necessary to *identify* it as a particular *asset* ("embedded memory is expensive")
- Mandatory shall be: device/module holds additional information to *maintain* its "life cycle" (e.g. HW and SW releases)
- Optional (1) should be: device/module provides parameters to store information about its *usage* (e.g. TAG, installation date, descriptor)
- Optional (2) should be: parameters that allow assigning the device/module status to various *history logs* (audit trail, MES, asset management, etc.)

- Only that information shall be stored that allows *easy replacement* of a device/module without the help of (SW)-tools
- *Permanent storage* within the device/module is required even with disconnected power. The profile guidelines do not specify where the storage takes place. A head station may act as a representative for modules (virtual or real modules).
- Data records with I&M parameters shall be readable (or write able if applicable) via *MS2 services (mandatory) and MS1 services (optional)* [1], [4], [5].
- The I&M information shall be *helpful for the user* (will he be able to define the necessary corrective actions for the field service personnel?) Individual manufacturer specific "information encryption" may not be good enough).
- Brandlabeling: It is possible for different companies to brandlabel devices with the same Ident Number (see 7.4.1). The focus with the I&M functions is to provide the best customer support possible. The "brandlabeling" company has to decide what kind of information is best for its customers and shall clarify with the original manufacturer.

1.2 Terms and Definitions

Asset Management	Asset management is the process by which assets and their assignments are recorded and tracked in a centralized database to ensure accurate organization of an asset inventory.
Audit Trail	The creation of a chronological record of system activities that is sufficient to enable the reconstruction, review, and examination of the sequence of environments and activities surrounding or leading to an operation, procedure, or an event in a transaction from its inception to its final results.
(Field)-Device	Component connected to a field bus that interacts directly with the process, usually remote I/O, sensors, actuators and alike. Inter-networking components are not Field Devices.
Head Station	Within a real or virtual modular (field)-device a module that plays the role of a supervisor for all the attached modules. Usually it contains the field bus connection.
Index	The third layer within the PROFIBUS DP's hierarchical addressing scheme: field bus node → slot → index
Load Domain	IEC 61158 defines procedures within the application layer to up- and download bulky data up to 64 Kbytes. These services are called "load domain". One of them is the "CALL_REQ", that supports sub indices up to 65535.
Manufacturing Execution Systems (MES)	MESs bring pervasive computerization to plant floors by providing such diverse functions as quality control, document management, and plant-floor dispatching. By capturing "live" information about setups, run times, throughput, and yields, managers are better able to measure constraints, identify bottlenecks, get a better understanding of capacity, and close the loop for production management, ensuring production per the plan.
Module	A physical unit within a modular field device. It is assigned to a particular slot of the PROFIBUS slot-addressing scheme. It may extend more than one slot.
Signature	An electronic signature based upon cryptographic methods of originator authentication, computed by using a set of rules and a set of parameters such that the identity of the signer and the integrity of the data can be verified [3]. Authenticity and data integrity for a given field bus device can be determined by the asset identification together with the signature.
Slot	The second layer within the PROFIBUS DP's hierarchical addressing scheme: field bus node → slot → index

<i>Sub-Index</i>	The fourth layer within the PROFIBUS DP's hierarchical addressing scheme: field bus node → slot → index → sub index (0 – 65535).
<i>TAG</i>	This is an alphanumeric label given to individual distributed control units and/or fieldbus devices so that the system can identify them. The label may indicate the functional role of the device and/or its location within the plant.
<i>Type Plate</i>	Or "Boiler Plate". Usually the label attached to any product identifying its manufacturer, type, production number and alike. The I&M functions for PROFIBUS devices are kind of an extended electronic type plate.

See IEC 61158 [1] for further terms and definitions.

1.3 Standards

IEC 61158	Digital data communications for measurement and control - Fieldbus for use in industrial control systems, Parts 2-6
IEC 61784	Communication profiles for field bus – profile sets for continuous and discrete manufacturing, type 3 (PROFIBUS)
IEC 61508	Functional safety of electrical/electronic/programmable electronic safety-related systems, Parts 1-7
ANSI/ISA-95.00.01-2000	Enterprise-Control System Integration, Part 1: Models and Terminology
ANSI/ISA-95.00.02-2001	Enterprise-Control System Integration, Part 2: Object Models Attributes
21 CFR 11	US Federal Food, Drug and Cosmetic Act: Title 21, Code of Federal Regulations, Part 11 – Electronic Records; electronic signatures
ISO/IEC 646:1991	Information technology -- ISO 7-bit coded character set for information interchange
ISO 8601:2000	Data elements and interchange formats -- Information interchange -- Representation of dates and times

2 I&M Index Space and Partitions

In order to avoid conflicts with many already installed PROFIBUS devices in the field and to save address space for operational parameters, this proposal follows the CALL_REQ service defined within IEC 61158-6. This service, part of the "Load Domain" Up/Download services, can be used within any module independent from any directory in a representative module (e.g. slot 0) of a device. It uses index 255 within any slot and opens a separate addressable sub index space. For I&M functions the sub index range from 65000 to 65199 is reserved. They are called IM_Index.

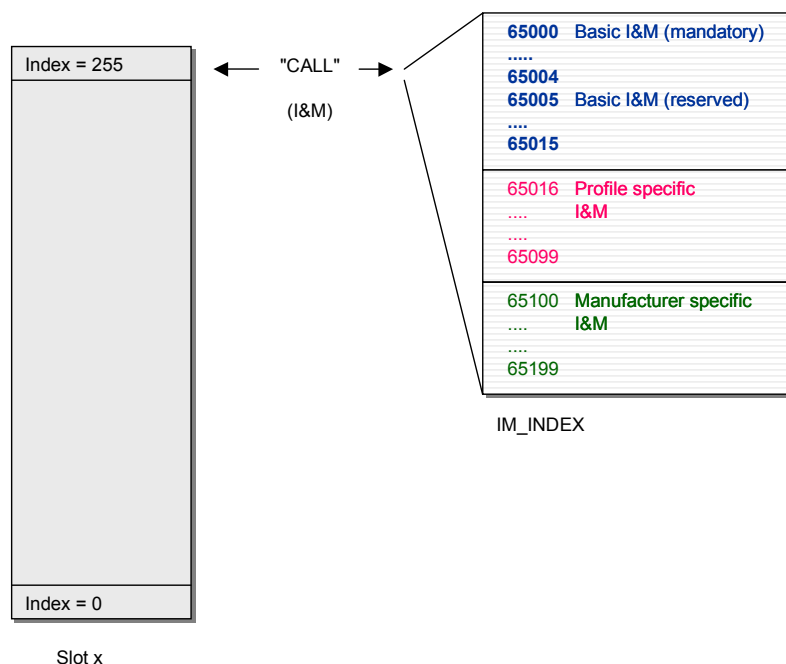


Figure 2-1 Sub Index Space for I&M Functions

The CALL_REQ service needs several header octets thus reducing the possible net data length to 236 bytes.

For I&M functions the following block of sub indices (IM_INDEX) will be used:

IM_INDEX	Usage
65000	I&M0
65001	I&M1
65002	I&M2
65003	I&M3
65004	I&M4
65005 ...65015	Reserved for additional general I&M functions
65016 ...65099	Profile specific I&M functions
65100 ...65199	Manufacturer specific I&M functions

3 I&M Parameters

3.1 Responsibilities

There are 3 responsibilities to provide or update the parameter information:

- The device's firmware (Initiator = "Firmware") is storing the appropriate information.
- Parameters marked with Initiator = "User" can be set at the customer's needs. The manufacturer shall preset them at production time to a certain default value.
- Parameters marked with Initiator = "Production" shall be set at production time by the manufacturer.

3.2 Parameter Definition

3.2.1 Header

Manufacturers may use the header for individual accompanying information. Third party tools shall ignore and not display this information. For a write access they shall use the same header information they got during a preceding read access.

3.2.2 MANUFACTURER_ID

The parameter MANUFACTURER_ID carries the ID for a specific manufacturer that had been assigned by the PROFIBUS business office. Default value is 0000h in case a manufacturer does not want to handle an ID.

This approach prefers IDs rather than company names. Over the years companies may change names or responsibilities for products so that a customer may not be able to find the right help. Using today's Internet, a dynamic list of IDs and its associated actual company names and support "pointers" seems to be more appropriate.

Name	Size	Data Type	Initiator	Action
MANUFACTURER_ID	2 Octets	Unsigned 16	Firmware	-
			User	-
			Production	e.g. "42"

The PROFIBUS community will arrange for a human readable list or table in XML on its web pages that can be downloaded by any engineering or service tool whenever it's connected to the Internet. Thus with the help of the MANUFACTURER_ID these tools may offer information about the manufacturer's company and information on how to reach customer support (phone numbers, etc.). See www.profibus.com for further details.

When connected to the Internet an even more comfortable support will be possible if the tool is able to interpret an "html query" format coming with the downloaded table. See chapter 3.2.3 for further details.

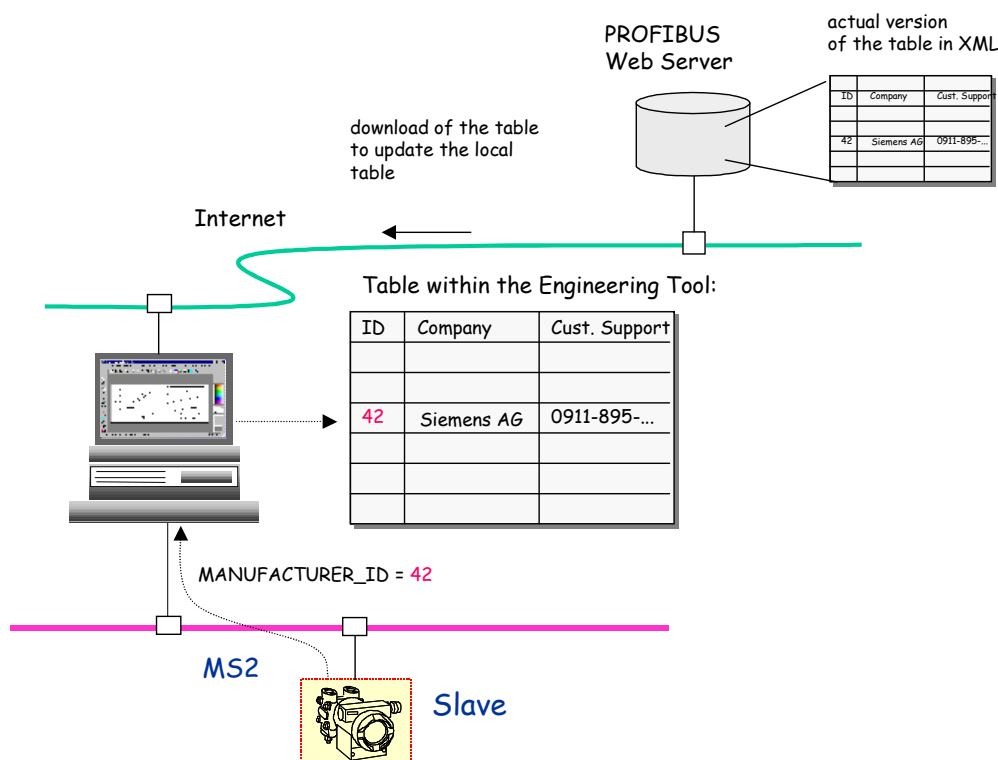


Figure 3-1 Referencing of Manufacturer IDs via Internet

3.2.3 ORDER_ID

This parameter contains the complete order number or at least a relevant part that allows unambiguous identification of the device/module within the manufacturer's web site. In contrast to [2] it comprise 20 octets. Unused octets shall be set "20h" (blank).

Name	Size	Data Type	Initiator	Action
ORDER_ID	20 Octets	Visible String	Firmware	-
			User	-
			Production	e. g. "3xy-0AE00-0AB0"

The MANUFACTURER_ID table (in XML format) may contain the URL of the search engine of the manufacturer's web site and the format of its HTML query with formal parameters that are marked with %..., e.g.

www.company.com/products/search/ordernumber/%ORDER_ID/language/%LNG

The tool reads the ORDER_ID out of the device/module and replaces the formal parameters by actual parameters. The HTML query may contain additional formal parameters such as language (%LNG) that can be replaced by the tool itself. The PROFIBUS web site will publish a list of known formal parameters and its semantics.

With the help of such an HTML query the tool may address the search engine of a particular manufacturer's web site and display the results. Web browser viewers to integrate in engineering or service tools are easily available on the market.

The PROFIBUS web site will publish detailed information about the data structures of HTML queries and other product support opportunities.

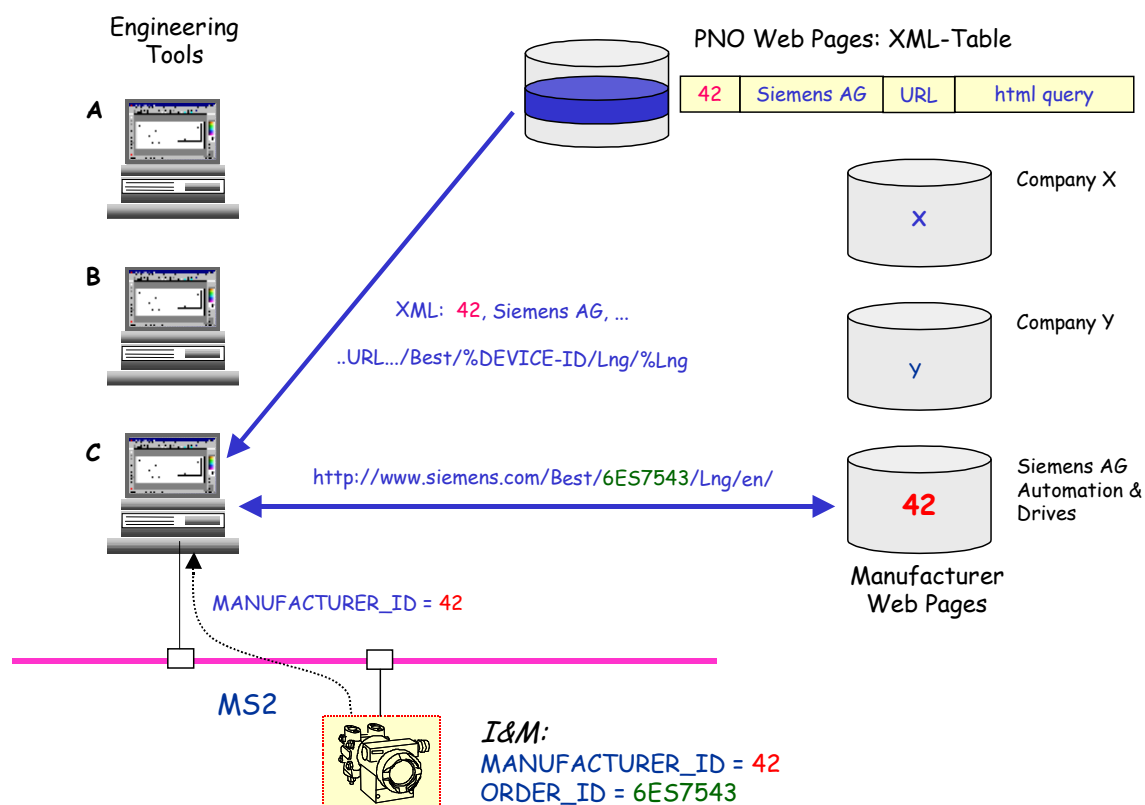


Figure 3-2 Approaching Product Information via HTML Queries

3.2.4 SERIAL_NUMBER

A serial number is a unique production number of the device manufacturer even for devices with the same hardware, software or firmware edition. Unused characters shall be set to 20h (blank).

Name	Size	Data Type	Initiator	Action
SERIAL_NUMBER	16 Octets	Visible String	Firmware	-
			User	-
			Production	Unique serial number

3.2.5 HARDWARE_REVISION

The content of this parameter characterizes the edition of the hardware only. Default value is 0000h. FFFFh indicates availability of profile specific information. Hint: individual revision information may be stored within the manufacturer specific parts of the I&M functions. However, this information can only be made available via EDD or DTM.

Name	Size	Data Type	Initiator	Action
HARDWARE_REVISION	2 Octets	Unsigned16	Firmware	-
			User	-
			Production	e. g. "0...65535"

3.2.6 SOFTWARE_REVISION

The content of this parameter characterizes the edition of the software or firmware of a device or module. The structure supports coarse and detailed differentiation that may be defined by the manufacturer: Vx.y.z. Any component that carries software shall present its software version even if the user may not be aware of it. V255.255.255 indicates the availability of profile specific information.

Hint: profile and/or manufacturer specific extensions are possible (see chapter 3.3 and 3.4).

Name	Size	Data Type	Initiator	Action
SOFTWARE_REVISION	4 Octets	1 Char, 3 Unsigned 8	Firmware	e.g. "V1.3.0"
			User	-
			Production	-

Detailed coding:

Octet	Data Type	Notation	Range
0	Char	Type Recognition	V *)
1	Unsigned 8	(x) Functional enhancement	0..255
2	Unsigned 8	(y) Bug fix	0..255
3	Unsigned 8	(z) Internal change (no impact on function)	0..255

*) Preference: V = officially released version or R = Revision (of a virtual or physically modular product).

Other defined characters are:

P = Prototype; U = Under Test (field test); T = Test Device

3.2.7 REV_COUNTER (Revision Counter)

A changed value of the REV_COUNTER parameter of a given module marks a change of hardware or of its parameters. At production time the counter will be set to 0. This value is reserved for the first installation and the first increment. Thus, the counter increments from 1... 65535, wrapping over back to 1 at the end.

Criteria to increment the counter:

- Module exchange
- Any write access of parameters within the module and its activation

To avoid unnecessary counter increments more tolerant and sophisticated criteria, meeting the minimum requirements of FDA (Food & Drug Administration) may be implemented manufacturer specifically (e.g. batch-ID, local parameter server, etc).

Exception: the REV_COUNTER within the representative of the whole modular device (slave) – usually assigned to slot 0 – shall be incremented if any module's REV_COUNTER is incremented.

Name	Size	Data Type	Initiator	Action
REV_COUNTER	2 Octets	Unsigned 16	Firmware	Increment (1...65535)
			User	-
			Production	0

3.2.8 PROFILE_ID

A module following a special profile may offer extended information (PROFILE_SPECIFIC_TYPE) about its function and/or sub devices, e.g. HART. Devices not following any PROFIBUS application profile shall use F600h. In this case the device is generic and may indicate its type via the parameter PROFILE_SPECIFIC_TYPE.

Name	Size	Data Type	Initiator	Action
PROFILE_ID	2 Octets	Unsigned 16	Firmware	(Optional)
			User	-
			Production	Specific profile ID, e.g. 5F00

The following table shows currently assigned PROFILE-IDs to its profiles:

3A00 – 3AFF	PROFIdrive
3B00 – 3BFF	Robots and Numeric Controls
3C00 – 3CFF	Display & Panel Devices
3D00 – 3DFF	Encoder
3E00 – 3EFF	PROFIsafe
3F00 – 3FFF	Fluid Power / Hydraulics
4900 – 49FF	Water Treatment
4B00 – 4BFF	Reserved
4C00 – 4CFF	Reserved
4E00 – 4EFF	Reserved
4F00 – 4FFF	Reserved
5A00 – 5AFF	SEMI – Devices
5B00 – 5BFF	Identification Systems
5C00 – 5CFF	Weighing & Dosage Systems
5D00 – 5DFF	Intelligent Pumps
5E00 – 5EFF	Low Voltage Switchgear
5F00 – 5FFF	Remote I/O for Process Control
6200 – 62FF	Laboratory Devices
6A00 – 6AFF	reserved
6B00 – 6BFF	reserved
6C00 – 6CFF	reserved
6D00 – 6DFF	reserved
6E00 – 6EFF	reserved
6F00 – 6FFF	reserved
9700 – 97FF	PA Devices (for Process Control)
F600 – F6FF	Generic device

Updated information will be available under www.profibus.com.

3.2.9 PROFILE_SPECIFIC_TYPE

In case a module follows a special profile (see table in chapter 3.2.8) this parameter offers information about the usage of its channels and/or sub devices. The module/channel information shall be according to the respective definitions of the application profile. Devices/modules not following any PROFIBUS application profile are generic and shall use F600h – F6FFh. In this case the parameter indicates the type of sensor or actuator ("device type"; coding defined in Appendix 7.1).

Name	Size	Data Type	Initiator	Action
PROFILE_SPECIFIC_TYPE	2 Octets	Unsigned 16	Firmware	Profile specific code
			User	-
			Production	(Optional)

Example for the usage: in case of PA-Devices (PROFILE_ID = 97xx) the data type of PROFILE_SPECIFIC_TYPE is to be represented as 2 unsigned8:

Octet 1 (MSB) = BLOCK OBJECT

Octet 2 (LSB) = PARENT CLASS

3.2.10 IM_VERSION

This parameter indicates the implemented version of the I&M functions.

Name	Size	Data Type	Initiator	Action
IM_VERSION	2 Octets	2 Unsigned 8	Firmware	-
			User	-
			Production	According to the version of the Profile Guidelines Part1 "I&M Functions".

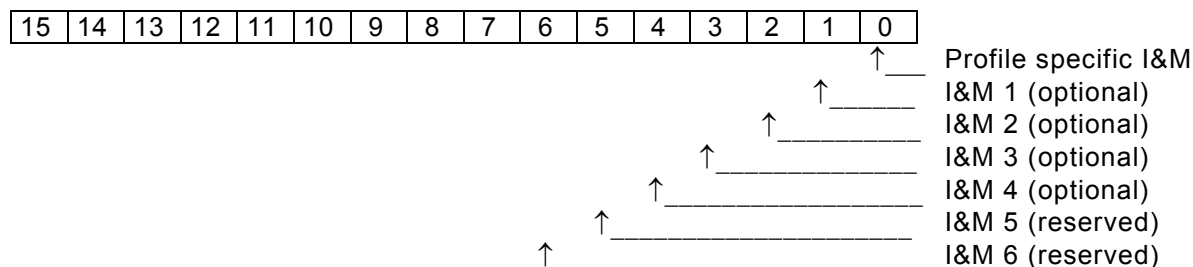
Octet 1 (MSB) = major version number, e.g. **1** of version 1.0

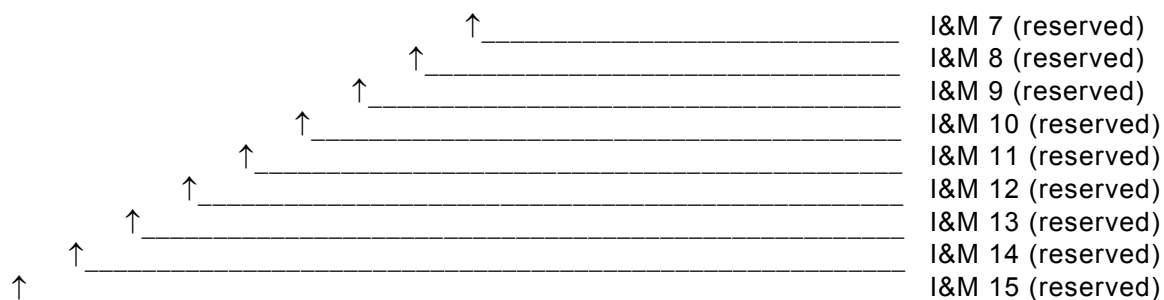
Octet 2 (LSB) = minor version number, e.g. **0** of version 1.0

3.2.11 IM_SUPPORTED (I&M directory)

This parameter indicates the availability of I&M records.

Name	Size	Data Type	Initiator	Action
IM_SUPPORTED	2 Octets	Unsigned16 (interpreted as Bit Array)	Firmware	-
			User	-
			Production	According to I&M implementation





3.2.12 TAG_FUNCTION

For each device or module within a plant a unique label is necessary for the identification of its function or task. This may be a standard symbolic tag ("AKZ") out of a list or any other type of label defined by a configuration tool. Unused characters shall be set to "20h" (blank).

Name	Size	Data Type	Initiator	Action
TAG_FUNCTION	32 Octets	Visible String	Firmware	-
			User	TAG entry
			Production	Filled with "20h" (blank)

3.2.13 TAG_LOCATION

For each device or module within a plant a unique label is necessary for the identification of its location. This may be a standard location tag ("OKZ") out of a list or any other type of label defined by a configuration tool. Unused characters shall be set to "20h" (blank).

Name	Size	Data Type	Initiator	Action
TAG_LOCATION	22 Octets	Visible String	Firmware	-
			User	TAG entry
			Production	Filled with "20h" (blank)

3.2.14 INSTALLATION_DATE

The parameter INSTALLATION_DATE indicates the date of installation or commissioning of a device or module. In contrast to [2] this profile suggests a fixed date format according to ISO 8601: YYYY-MM-DD, where YYYY is the year in the usual Gregorian calendar, MM is the month of the year between 01 (January) and 12 (December), and DD is the day of the month between 01 and 31.

Time information shall be added: hh:mm, where hh is the hour of the day (0...23) and mm the minutes of the hour (0...59). Time and date shall be separated by a blank character.

For example, the fourth day of February in the year 1995 is written in the standard notation as:

1995-02-04 16:23

Unused characters shall be set to "20h" (blank).

Name	Size	Data Type	Initiator	Action
INSTALLATION_DATE	16 Octets	Visible String	Firmware	-

			User	Installation date
			Production	Filled with "20h" (blank)

3.2.15 DESCRIPTOR (Comment)

This comment field DESCRIPTOR allows customers to store any individual additional information and annotation. One source can be the tag list. Unused characters shall be set to "20h" (blank).

Name	Size	Data Type	Initiator	Action
DESCRIPTOR	54 Octets	Visible String	Firmware	-
			User	Comment
			Production	Filled with "20h" (blank)

3.2.16 SIGNATURE (Security)

The parameter SIGNATURE allows parameterization tools to store a "security" code as a reference for a particular parameterization session and audit trail tools to retrieve the code for integrity checks. Together with the "asset identification" consisting of MANUFACTURER_ID, ORDER_ID, and SERIAL_NUMBER the SIGNATURE parameter allows unambiguousness of the session data. Unused characters shall be set to "20h" (blank).

Name	Size	Data Type	Initiator	Action
SIGNATURE	54 Octets	OctetString	Firmware	-
			User	Via parameterization tools
			Production	Filled with "00h"

Background: security, safety and data integrity are becoming more and more important within the automation business. Thus regulations like 21 CFR 11 are defining "rules for the use of secure, computer-generated, time-stamped audit trails to independently record the date and time of operator entries and actions that create, modify, or delete electronic records. Record changes shall not obscure previously recorded information. Such audit trail documentation shall be retained for a period at least as long as that required for the subject electronic records and shall be available for agency review and copying".

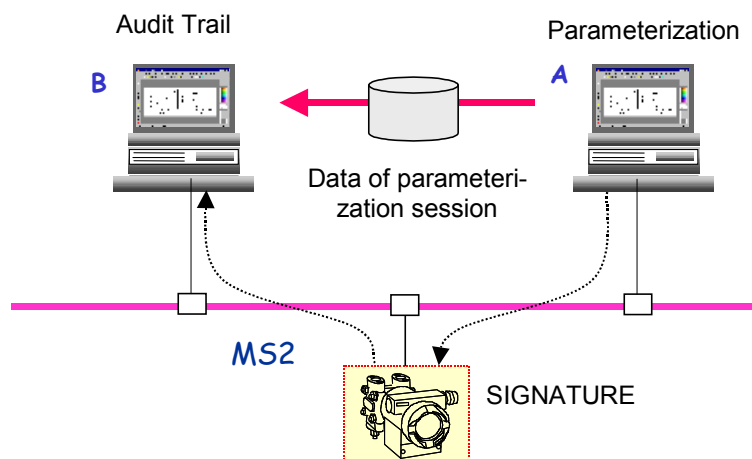


Figure 3-3 Use Case Parameterization Session

3.3 Profile specific Information

A profile working group may define profile specific information within the IM_INDEX range 65016 to 65099. Its availability within a device/module shall be indicated via Bit 0 = 1 of the parameter IM_SUPPORTED. The structure and data types of the information shall be defined via XML schema, downloadable from the PNO web sites. The reference for the profile is contained in the parameter PROFILE_ID. The length of data sets is not limited to 64 Octets.

3.4 Manufacturer specific Information

Device manufacturers can use the IM_INDEX range 65100 to 65199 to store and retrieve e.g. liability critical information that only can be accessed by their own tools (e.g. DTM). Usually this area will be protected by manufacturer defined password mechanisms. The header of a data record may be used to secure the access. The length of data sets is not limited to 64 Octets.

3.5 Data Types

Unsigned8

An Unsigned8 represents an unsigned number depicted by 8 bits

Coding	Data type	Range of values	Resolution	Length
5	Unsigned8	$0 \leq i \leq 255$	1	1 octet

Bit	7	6	5	4	3	2	1	0
Octet 1	2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0

Unsigned16

An Unsigned16 represents an unsigned number depicted by 16 bits.

Coding	Data type	Range of values	Resolution	Length
6	Unsigned16	$0 \leq i \leq 65535$	1	2 octets

Bit	7	6	5	4	3	2	1	0
Octet 1	2^{15}	2^{14}	2^{13}	2^{12}	2^{11}	2^{10}	2^9	2^8
Octet 2	2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0

VisibleString

This data type is defined as the ISO 646 string type. Characters are based on 7 Bit ASCII

Coding	Data type	Range of values	Resolution	Length
9	VisibleString	Refer to ISO 646	-	Variable

Bit	7	6	5	4	3	2	1	0
Octet 1	1st character							
Octet 2	2nd character							
...								
...								
Octet n	nth character							

OctetString

An OctetString is an ordered sequence of Octets, numbered from 1 to n

Coding	Data type	Range of values	Resolution	Length
10	OctetString	-	-	Variable

Bit	7	6	5	4	3	2	1	0
Octet 1	1. Byte							
Octet 2	2. Byte							
...								
...								
Octet n	n. Byte							

4 I&M Coding of Records

In order to transport the I&M parameters across the PROFIBUS network via MS1 (optional) or MS2 (mandatory) services they shall be packed into records. Records can be read and/or written. Device manufacturers usually like to protect the way parameters are written into the device. Thus it is the intention of this profile to standardize the read record structures only so they can be read out by third party tools, e.g. handheld devices. The headers may be used during write sequences for manufacturer specific purposes, e.g. directory information, write protection mechanisms like passwords and alike. The length of data sets is limited to 64 Octets.

The I&M parameters shall be partitioned into five different kinds of read data sets, each limited to 64 Octets:

4.1 I&M0 (mandatory)

Content	Size	Coding (H)
<i>Header</i>		
Manufacturer specific	10 Octets	Manufacturer specific
<i>I&M Block</i>		
MANUFACTURER_ID	2 Octets	Chapter 3.2.1
ORDER_ID	20 Octets	Chapter 3.2.3
SERIAL_NUMBER	16 Octets	Chapter 3.2.4
HARDWARE_REVISION	2 Octets	Chapter 3.2.5
SOFTWARE_REVISION	4 Octets	Chapter 3.2.6
REVISION_COUNTER	2 Octets	Chapter 3.2.7
PROFILE_ID	2 Octets	Chapter 3.2.8
PROFILE_SPECIFIC_TYPE	2 Octets	Chapter 3.2.9
IM_VERSION	2 Octets	Chapter 3.2.10
IM_SUPPORTED	2 Octets	Chapter 3.2.11

Length of I&M block / data set = 54 / 64 Octets

4.2 I&M1 (optional)

Content	Size	Coding (H)
<i>Header</i>		
Manufacturer specific	10 Octets	Manufacturer specific
<i>I&M Block</i>		
TAG_FUNCTION	32 Octets	Chapter 3.2.12
TAG_LOCATION	22 Octets	Chapter 3.2.13

Length of I&M block / data set = 54 / 64 Octets

4.3 I&M2 (optional)

Content	Size	Coding (H)
<i>Header</i>		
Manufacturer specific	10 Octets	Manufacturer specific
<i>I&M Block</i>		
INSTALLATION_DATE	16 Octets	Chapter 3.2.14
RESERVED	38 Octets	

Length of I&M block / data set = 54 / 64 Octets

4.4 I&M3 (optional)

Content	Size	Coding (H)
<i>Header</i>		
Manufacturer specific	10 Octets	Manufacturer specific
<i>I&M Block</i>		
DESCRIPTOR	54 Octets	Chapter 3.2.15

Length of I&M block / data set = 54 / 64 Octets

4.5 I&M4 (optional)

Content	Size	Coding (H)
<i>Header</i>		
Manufacturer specific	10 Octets	Manufacturer specific
<i>I&M Block</i>		
SIGNATURE	54 Octets	Chapter 3.2.16

Length of I&M block / data set = 54 / 64 Octets

5 IEC 61158 Standard Call Procedure

5.1 Write Call (I&M Request)

The call mechanisms in use for the I&M functions are defined within IEC 61158-6, chapter 6.1 [1].

The data structure of a Write Call may comprise a DP-V1 header, a Call header and the body.

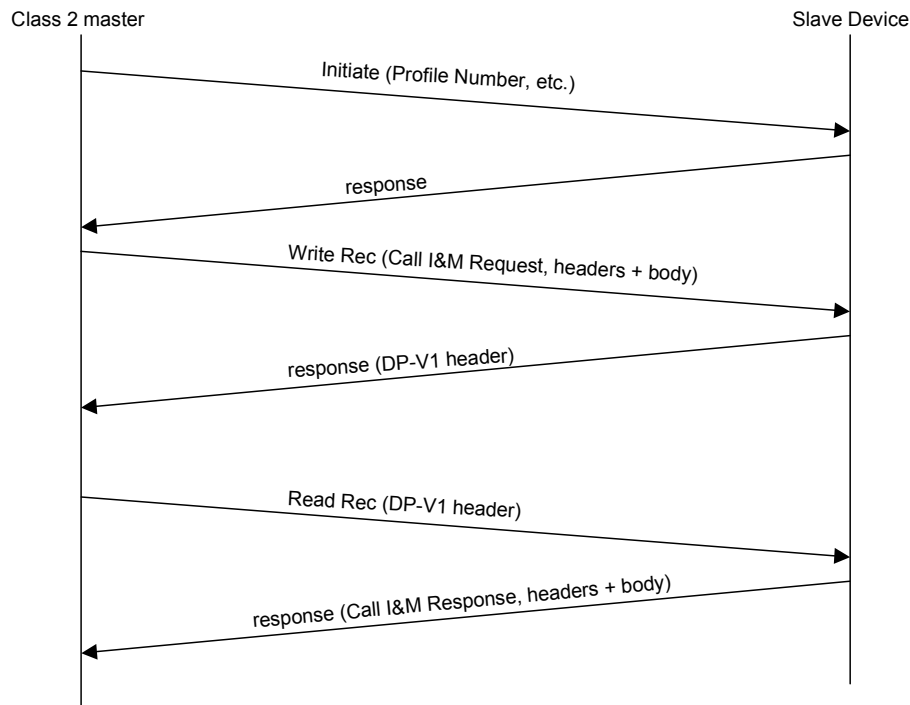
Content of a Write Call	Size	Coding	Notes	
FUNCTION_NUM	1 Octet	5Fh	Indicates "write", fix	DP-V1 Header
Slot number	1 Octet	0 ... 255	variable	
Index	1 Octet	255	fix	
Length of net data	1 Octet	68 (with body)	Length of I&M data set	
EX-TENDED_FUNCTION_NUM	1 Octet	08h	Indicates "CALL", fix	Call Header
reserved	1 Octet	00h	fix	
IM_INDEX	2 Octets	65000 ... 65199	Reserved for I&M	
IM_FUNCTION	64 Octets	Acc. definition	e.g. I&M3 descriptor	Body*)

*) In case of "Read Record" there is no body with the Write Call.

5.2 Read Call (I&M Response)

Content of a Read Call	Size	Coding	Notes	
FUNCTION_NUM	1 Octet	5Eh	Indicates "read", fix	DP-V1 Header
Slot number	1 Octet	0 ... 255	variable	
Index	1 Octet	255	fix	
Length of net data	1 Octet	68 (with body)	Length of I&M data set	
EX-TENDED_FUNCTION_NUM	1 Octet	08h	Indicates "CALL", fix	Call Header
reserved	1 Octet	00h	fix	
IM_INDEX	2 Octets	65000 ... 65199	Reserved for I&M	
IM_FUNCTION	64 Octets	Acc. definition	e.g. I&M3 descriptor	Body*)

*) In case of "Write Record" there is no body with the Read Call.



The sequence for writing and reading I&M data sets is the same. In order to only read an I&M data set the body in the Write Rec telegram remains empty.

5.3 Access Coordination

Currently there is no standardized common procedure to govern multi MS2 master accesses and/or MS1/MS2 accesses. It is up to the PROFIBUS device (slave) to establish a proper locking. Here some implementation hints:

- Parameters should be read back after a write operation
- The device shall allocate individual buffers for every SAP of a master device logged in. It is responsible for interlocking master devices in a proper way.
- In case a login will be rejected the device shall send a proper error message, e.g. "access denied" if index 255 is not implemented.
- A watchdog shall supervise any acyclic connection. Its time is dependent on the baud rate (see IEC 61158 for details)
- No password should be needed for I&M functions
- The correct procedure for Write Records is a 2-phase commit transaction.
- Currently several PROFIBUS Working Groups are discussing proposals for access coordination including 2-phase commit transaction, HART master "handles", and alike. These guidelines will be updated once an agreement is achieved.

6 GSD Keywords

The availability of standard I&M functions within a PROFIBUS slave will be indicated via a new keyword

Ident_Maintenance_supp (O)

Type: Boolean (1:true)

It can be used for the whole device or module specific.

7 Appendix

7.1 Coding of IDs and Types

XML based files will be published on the PROFIBUS web site in machine and human readable form by mid of 2003.

7.2 Applicable Documents

- [1] IEC 61158 by www.iec.ch
- [2] PROFIBUS Application Profile "PA Devices", V3.0, Order-No: 3.042
- [3] Food & Drug Administration: 21 CFR Part 11; Section 11.3, Definitions
- [4] "PROFIBUS System Description - Technology and Application ", October 2002, free download from www.profibus.com or PNO-Order-No: 4.002
- [5] Manfred Popp: "The New Rapid Way to PROFIBUS DP", November 2002, PNO-Order-No: 4.072

7.3 Abbreviations

DTM	Device Type Manager
GSD	Generic Station Description ("Geräte-Stammdaten")
IEC	International Electrotechnical Commission
I/O	Input / Output
ISO	International Standards Organization
MES	Manufacturing Execution Systems
MS1	Acyclic communication services between a master class 1, e.g. controller and its slave
MS2	Acyclic communication services between a master class 2, e.g. laptop and its slave
SAP	Service Access Point
URL	Universal Resource Locator (world wide web)

7.4 Rules for the Assignment and the Usage of PROFIBUS IDs

The PROFIBUS business office is managing a data base of general IDs within a range from 0000h ... FFFFh which are used for the following purposes.

7.4.1 Manufacturer specific ID

In this case the ID is a unique device identification that is hard-coded within the slave and that is exchanged with the master in case of a parameterization telegram (Prm) or a diagnosis message. The IEC 61158 defines an "Ident Number" for this usage: "The Ident Number specifies a DP master (class1) or DP slave device type" (IEC 61158-6, 3.4.4.24) [1].

The "Ident Number" is part of the GSD file name and also is keyword within the GSD file. Thus ensuring that a configuration tool is using the right GSD file for a particular device and the master and slave configuration fit together. After a device exchange, data communication may resume after verification of the "Ident Number".

Upon application the PROFIBUS business office assigns either single "Ident Numbers" or a range of numbers to a particular company. Single numbers are assigned from the lower end (0000h) consecutively.

7.4.2 Profile specific ID

In cases where the device of a manufacturer A should be replaceable by an equivalent device of a manufacturer B the above mentioned administration of "Ident Numbers" is no longer sufficient. Therefore the PROFIBUS business office is assigning number ranges to dedicated device types (profile specific IDs) in combination with certain "Profile GSDs". Profiles using this methodology are e.g. "PA Devices" and "PROFIdrive".

Usually these slave devices are designed to communicate with a master class 2 application (e.g. profile application or profile DTM). In order to ensure a master application is communicating with an appropriate slave it is sending a profile specific ID during the establishment of the connection (MS2 Initiate Service). The slave may answer with the same profile specific ID (if it is supporting this profile), with another ID (if it is supporting another profile) or with "0000h" if it is not supporting any profile.

For this usage the IEC 61158 defines a "Profile_Ident_Number": "MS2 user parameter class specification" (IEC 61158-5, 8.2.5.2.8) [1].

7.4.3 PROFILE_ID (I&M Functions)

The I&M functions provide device information (asset) to identify a particular device (or a module in case of a modular device) in a non-ambiguous manner by indicating the manufacturer's name, the order number and a serial number. Additional information is provided by software and hardware version numbers.

Besides this basic information devices following a certain profile are enabled to provide more detailed profile specific information (e.g. transponder and weighing systems). A "PROFILE_ID" is used to reference these extended definitions (see chapter 3.2.8).

The existence of a PROFILE_ID indicates the availability of a PROFIBUS Application Profile with additional information about the functionality of the device via state charts, sequence charts, parameters and alike.

The PROFILE_ID may be interpreted by tools and used to download profile specific XML schemas from the PROFIBUS Web-Server and thus provide profile individual displays of I&M functions (dynamic adaptation).

7.4.4 Recommendations

At best one and the same PROFIBUS ID can be applied for all three use cases ("Ident number", Profile_Ident_Number and PROFILE_ID). At least Profile_Ident_Number and PROFILE_ID shall be identical.

Profile Subclasses

An Application Profile Working Group is responsible for the subdivision of the PROFILE_ID range into subclasses. In case of low voltage switchgears these could be switchgear, control-gear, circuit-breaker, motor-starter, etc.

Multiple Profile Devices

The main purpose of a device is dominant for the choice of the PROFILE_ID. Example: a standard motor-starter and a PROFIsafe motor-starter: the profile choice shall be motor-starter. It is the working group's responsibility to define extended information within the profile specific part.

With the help of different "Profile_Ident_Numbers" during the establishment of the connection (Initiate) different tasks between a master application and its slave may be executed. Example: standard parameterization of the motor-starter with "Profile_Ident_Number" 5E00h and PROFIsafe parameterization with 3E00h.

Versioning

Versioning via "Ident number", "Profile_Ident_Number" and "PROFILE_ID" shall be avoided.

It is highly recommended to achieve upward compatibility from one profile version to the others. At least device exchange without impact on the user program in the controller shall be a major goal during the development of the profile. Reconfiguration and Reparameterization may be unavoidable (MS0, MS1).

In case of MS1 and MS2 connections either profile specific (e.g. Profile-DTM) or manufacturer specific I&M functions (e.g. individual DTM) may be used for version information.

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